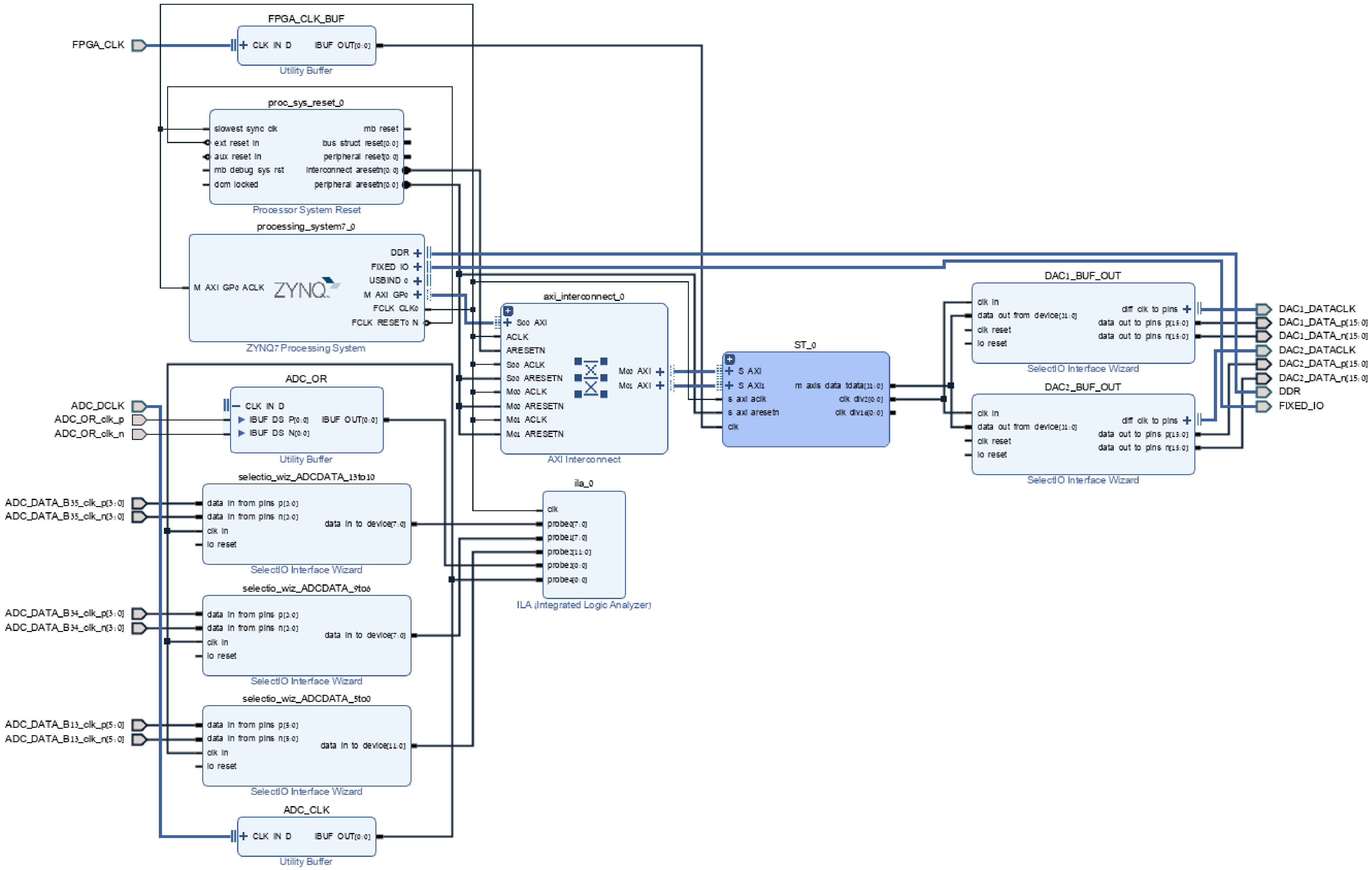
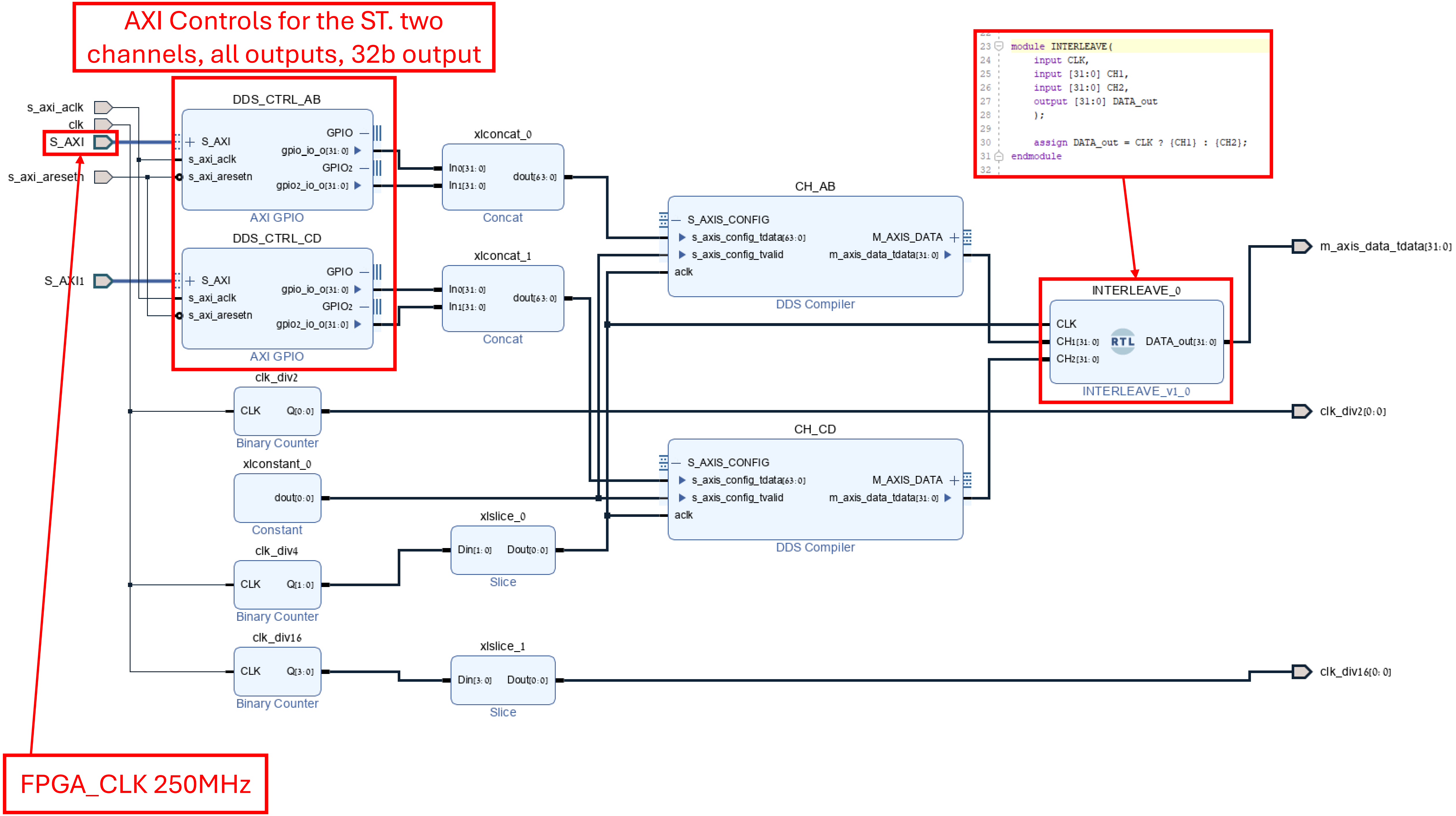
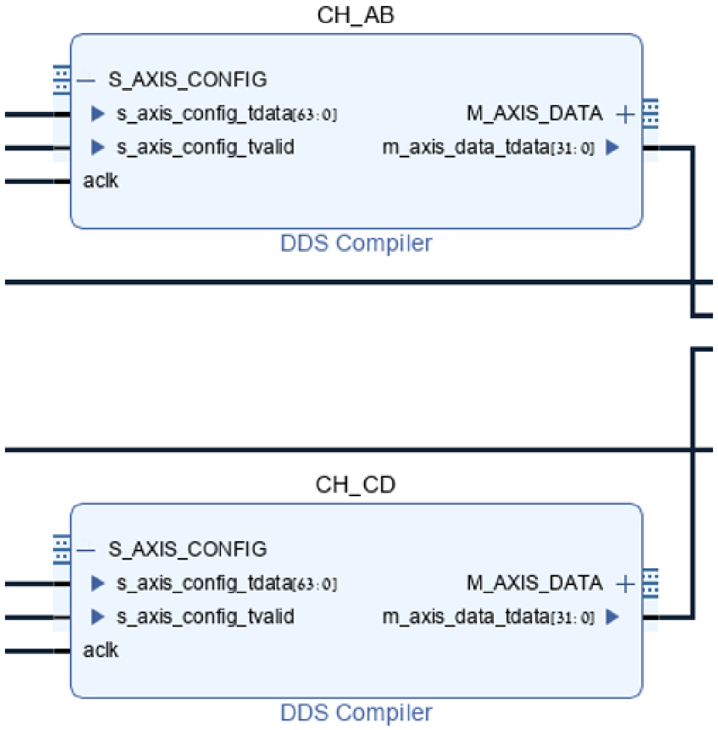
ADC and Single Tone Vivado BD Breakdown

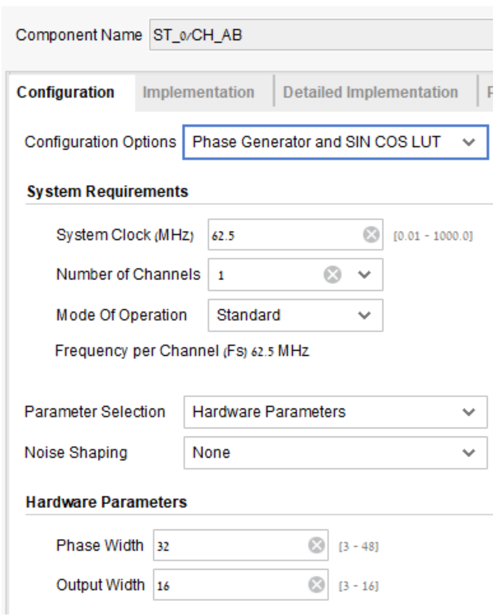


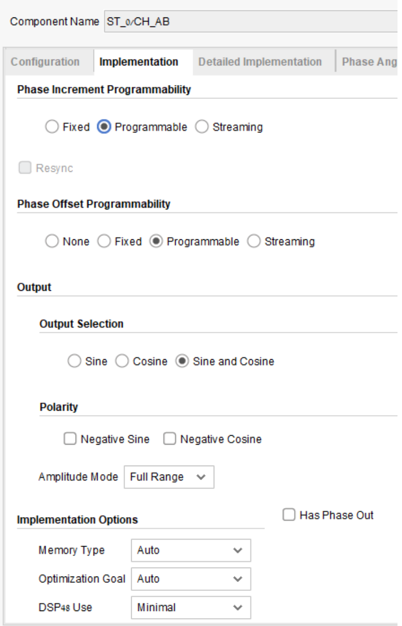
FPGA_CLK
FPGA_CLK_BUF
CLK IN D
IBUF OUT[0:0]
Utility Buffer
proc_sys_reset_0
slowest sync clk
ext reset in
aux reset in
mb debug sys rst
dcm locked
mb reset
bus struct reset[0:0]
peripheral reset[0:0]
interconnect aresetn[0:0]
peripheral aresetn[0:0]
Processor System Reset
processing_system7_0
M AXI GP0 ACLK
ZYNQ
DDR
FIXED IO
USBIND 0
M AXI GP0
FCLK CLK0
FCLK RESET0 N
ZYNQ7 Processing System
axi_interconnect_0
S00 AXI
ACLK
ARESETN
S00 ACLK
S00 ARESETN
M00 ACLK
M00 ARESETN
M01 ACLK
M01 ARESETN
M00 AXI
M01 AXI
AXI Interconnect
ST_0
S AXI
S AXI1
s axi aclk
s axi aresetn
clk
m axis data tdata[31:0]
clk div2[0:0]
clk div16[0:0]
DAC1_BUF_OUT
clk in
data out from device[31:0]
clk reset
io reset
diff clk to pins
data out to pins p[15:0]
data out to pins n[15:0]
SelectIO Interface Wizard
DAC2_BUF_OUT
clk in
data out from device[31:0]
clk reset
io reset
diff clk to pins
data out to pins p[15:0]
data out to pins n[15:0]
SelectIO Interface Wizard
DAC1_DATACLK
DAC1_DATA_p[15:0]
DAC1_DATA_n[15:0]
DAC2_DATACLK
DAC2_DATA_p[15:0]
DAC2_DATA_n[15:0]
DDR
FIXED_IO
ADC_DCLK
ADC_OR_clk_p
ADC_OR_clk_n
ADC_OR
CLK IN D
IBUF DS P[0:0]
IBUF DS N[0:0]
IBUF OUT[0:0]
Utility Buffer
ila_0
clk
probe0[7:0]
probe1[7:0]
probe2[11:0]
probe3[0:0]
probe4[0:0]
ILA (Integrated Logic Analyzer)
ADC_DATA_B35_clk_p[3:0]
ADC_DATA_B35_clk_n[3:0]
selectio_wiz_ADCDATA_13to10
data in from pins p[3:0]
data in from pins n[3:0]
clk in
io reset
data in to device[7:0]
SelectIO Interface Wizard
ADC_DATA_B34_clk_p[3:0]
ADC_DATA_B34_clk_n[3:0]
selectio_wiz_ADCDATA_9to6
data in from pins p[3:0]
data in from pins n[3:0]
clk in
io reset
data in to device[7:0]
SelectIO Interface Wizard
ADC_DATA_B13_clk_p[5:0]
ADC_DATA_B13_clk_n[5:0]
selectio_wiz_ADCDATA_5to0
data in from pins p[5:0]
data in from pins n[5:0]
clk in
io reset
data in to device[11:0]
SelectIO Interface Wizard
ADC_CLK
CLK IN D
IBUF OUT[0:0]
Utility Buffer

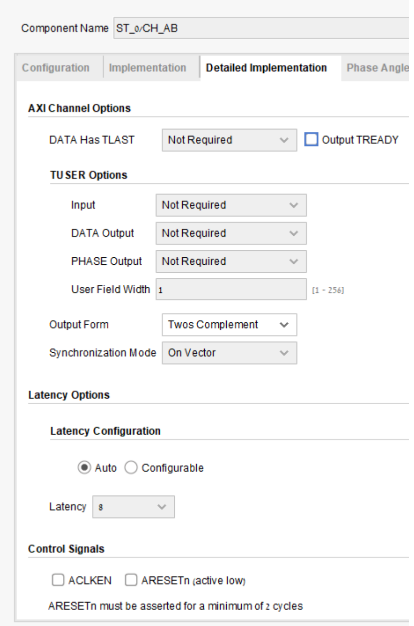


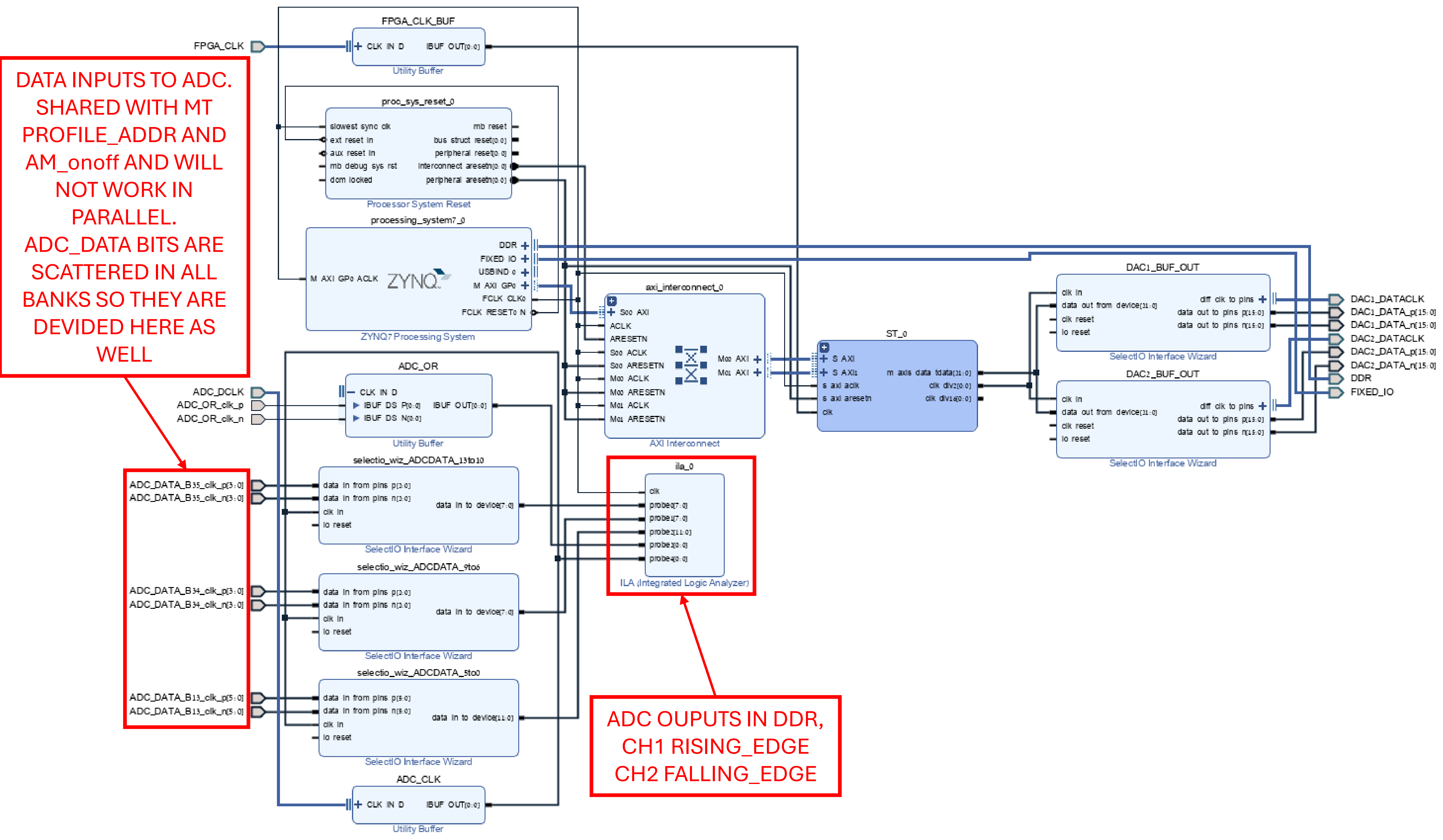
AXI Controls for the ST. two channels, all outputs, 32b output
FPGA_CLK 250MHz
s_axi_aclk
clk
S_AXI
s_axi_aresetn
S_AXI1
DDS_CTRL_AB
S_AXI
s_axi_aclk
s_axi_aresetn
GPIO
gpio_io_o[31:0]
GPIO2
gpio2_io_o[31:0]
AXI GPIO
DDS_CTRL_CD
S_AXI
s_axi_aclk
s_axi_aresetn
GPIO
gpio_io_o[31:0]
GPIO2
gpio2_io_o[31:0]
AXI GPIO
xlconcat_0
In0[31:0]
In1[31:0]
dout[63:0]
Concat
xlconcat_1
In0[31:0]
In1[31:0]
dout[63:0]
Concat
CH_AB
S_AXIS_CONFIG
s_axis_config_tdata[63:0]
s_axis_config_tvalid
aclk
M_AXIS_DATA
m_axis_data_tdata[31:0]
DDS Compiler
CH_CD
S_AXIS_CONFIG
s_axis_config_tdata[63:0]
s_axis_config_tvalid
aclk
M_AXIS_DATA
m_axis_data_tdata[31:0]
DDS Compiler
INTERLEAVE_0
CLK
CH1[31:0]
CH2[31:0]
RTL
DATA_out[31:0]
INTERLEAVE_v1_0
m_axis_data_tdata[31:0]
clk_div2[0:0]
clk_div16[0:0]
clk_div2
CLK
Q[0:0]
Binary Counter
xlconstant_0
dout[0:0]
Constant
clk_div4
CLK
Q[1:0]
Binary Counter
xlslice_0
Din[1:0]
Dout[0:0]
Slice
clk_div16
CLK
Q[3:0]
Binary Counter
xlslice_1
Din[3:0]
Dout[0:0]
Slice
module INTERLEAVE(
input CLK,
input [31:0] CH1,
input [31:0] CH2,
output [31:0] DATA_out
);
assign DATA_out = CLK ? {CH1} : {CH2};
endmodule


CH_AB
S_AXIS_CONFIG
s_axis_config_tdata[63:0]
s_axis_config_tvalid
aclk
M_AXIS_DATA
m_axis_data_tdata[31:0]
DDS Compiler
CH_CD
S_AXIS_CONFIG
s_axis_config_tdata[63:0]
s_axis_config_tvalid
aclk
M_AXIS_DATA
m_axis_data_tdata[31:0]
DDS Compiler


Component Name ST_0/CH_AB
Configuration
Implementation
Detailed Implementation
Configuration Options Phase Generator and SIN COS LUT
System Requirements
System Clock (MHz) 62.5 [0.01 - 1000.0]
Number of Channels 1
Mode Of Operation Standard
Frequency per Channel (Fs) 62.5 MHz
Parameter Selection Hardware Parameters
Noise Shaping None
Hardware Parameters
Phase Width 32 [3 - 48]
Output Width 16 [3 - 16]


Component Name ST_0/CH_AB
Configuration
Implementation
Detailed Implementation
Phase Ang
Phase Increment Programmability
Fixed
Programmable
Streaming
Resync
Phase Offset Programmability
None
Fixed
Programmable
Streaming
Output
Output Selection
Sine
Cosine
Sine and Cosine
Polarity
Negative Sine
Negative Cosine
Amplitude Mode Full Range
Has Phase Out
Implementation Options
Memory Type Auto
Optimization Goal Auto
DSP48 Use Minimal


Component Name ST_0/CH_AB
Configuration
Implementation
Detailed Implementation
Phase Angle
AXI Channel Options
DATA Has TLAST Not Required
Output TREADY
TUSER Options
Input Not Required
DATA Output Not Required
PHASE Output Not Required
User Field Width 1 [1 - 256]
Output Form Twos Complement
Synchronization Mode On Vector
Latency Options
Latency Configuration
Auto
Configurable
Latency 8
Control Signals
ACLKEN
ARESETn (active low)
ARESETn must be asserted for a minimum of 2 cycles



DATA INPUTS TO ADC. SHARED WITH MT PROFILE_ADDR AND AM_onoff AND WILL NOT WORK IN PARALLEL. ADC_DATA BITS ARE SCATTERED IN ALL BANKS SO THEY ARE DEVIDED HERE AS WELL
ADC OUPUTS IN DDR, CH1 RISING_EDGE CH2 FALLING_EDGE
FPGA_CLK
FPGA_CLK_BUF
Utility Buffer
proc_sys_reset_0
Processor System Reset
processing_system7_0
ZYNQ7 Processing System
axi_interconnect_0
AXI Interconnect
ST_0
DAC1_BUF_OUT
DAC2_BUF_OUT
SelectIO Interface Wizard
ADC_OR
ADC_DCLK
ADC_OR_clk_p
ADC_OR_clk_n
selectio_wiz_ADCDATA_13to10
selectio_wiz_ADCDATA_9to6
selectio_wiz_ADCDATA_5to0
ADC_CLK
ila_0
ILA (Integrated Logic Analyzer)
ADC_DATA_B35_clk_p[3:0]
ADC_DATA_B35_clk_n[3:0]
ADC_DATA_B34_clk_p[3:0]
ADC_DATA_B34_clk_n[3:0]
ADC_DATA_B13_clk_p[5:0]
ADC_DATA_B13_clk_n[5:0]
DAC1_DATACLK
DAC1_DATA_p[15:0]
DAC1_DATA_n[15:0]
DAC2_DATACLK
DAC2_DATA_p[15:0]
DAC2_DATA_n[15:0]
DDR
FIXED_IO